

Global Attitudes

Workshop

Peer Group Analysis with Pew Spring 2024 Data

Why Work Together?

The philosophy behind today's session



Explaining = understanding

You don't truly understand something until you can teach it. When an experienced student explains code line by line, they deepen their own knowledge — not just their partner's.



Learning by doing

Beginners learn by typing, not watching. Seeing code run on your own screen — and fixing your own errors — is irreplaceable. Reading along while someone else types doesn't stick.



Discussion drives insight

The best findings come from disagreement. When two people interpret a chart differently, they have to think harder. That friction is where learning happens.

What Are We Working With?

41,503

respondents

653

variables

35

countries

11

in our subset

Today's focus: European countries + Canada

France · Germany · Greece · Hungary · Italy · Netherlands · Poland · Spain · Sweden · United Kingdom · Canada

Topics: democracy satisfaction · economic pessimism · religiosity · social tolerance · support for autocracy

One Key Concept Before We Start

Survey weights — why every number in this dataset needs a weight

The problem

Not everyone is equally likely to respond to a survey. Younger men, rural residents, and people with less education respond less often — so the raw sample over-represents urban, educated, older women.

If we ignore this, our estimates reflect who answered, not the actual population.

The solution

Each respondent carries a weight — a number that says 'this person represents X people in the real population.'

weight = 2.0 → count twice

weight = 0.5 → count half

The `srvyr` package applies weights automatically once you create a survey design object with `as_survey_design()`.

Working With the Quarto File

1

Open your project

Make sure your .Rproj file is open. Top-right corner of RStudio shows the project name.

2

Open the .qmd file

The document has all instructions, hints, and code built in. Read the Mission box before touching any code.

3

Run the setup chunk first

Click ► on the 'load-packages' chunk. Then run 'load-data' and 'clean-data'. Do this once per session.

4

Work exercise by exercise

Run pre-written chunks with ► · Read the output · Then write the ✎ blank sections.

5

Write your interpretation

Type 2–3 sentences into the 💬 box after the code works. This is mandatory — not optional.

Running Code — Four Things to Know



One chunk at a time

Click the green ► or press Ctrl+Shift+Enter. Output appears immediately below the chunk. You do NOT need to click Render to see results.



Run All Chunks Above

If you see 'object not found', the chunk that creates it hasn't run yet. Use the ↓ button to catch up the entire environment in one click.



One line at a time

Ctrl+Enter runs only the line your cursor is on — perfect for testing a small change without rerunning the whole chunk.



Use the Console for experiments

The Console (bottom-left) is your scratch pad. Try things there first; only paste working code back into the document.

When Things Go Wrong

Three errors you will definitely see — and exactly how to fix them

Error: object 'pew_svy' not found

Fix: The setup chunk hasn't run yet. Click ↓ Run All Chunks Above on the current chunk.

Error: could not find function "survey_mean"

Fix: The srvyr package isn't loaded. Re-run the load-packages chunk at the top of the document.

Error: unexpected symbol / unexpected ','

Fix: Look at the line number in the error. Usually a missing comma, an unclosed parenthesis, or a |> on the wrong line.

Two Roles. One Group.

How this session works



Navigator

The person with more R experience

- Explain every line before it's typed
- Never grab the keyboard
- Ask questions, don't give answers
- Let the Driver struggle — that's learning
- If they're stuck, give a hint not a solution



Driver

The person doing the typing

- You control the keyboard
- Don't type anything you don't understand
- Ask before running — not after
- Read errors out loud, don't panic
- Your confusion is useful — voice it

90 Minutes — How It's Structured

0–10 min	Setup	Run the data chunks together. Navigator explains each line.
10–45 min	Exercises 1–3	Guided exercises — most code is pre-written. Navigator explains, Driver types.
~45 min	 Switch roles	Stop. Switch Navigator and Driver. The new Navigator recaps the pipeline.
45–75 min	Exercises 4–6	Less scaffolding. Exercise 6 is fully open — your group chooses the question.
75–90 min	Share & discuss	Each group shares one finding. 2–3 minutes per group.

Each Exercise Has the Same Four Parts

 1

Mission

One paragraph in plain English. The sociological question and why it matters. Read this before opening RStudio.

 2

Code — mostly pre-written

The code runs step by step. Navigator explains each line before the Driver types. Some sections are blank () — those are yours.

 3

Your interpretation

2–3 sentences written by the group. The questions point you at what to look for — answer them in your own words.

 4

Go further (optional)

A harder extension for groups that finish early. No scaffolding — you design the approach.

Ex 1

Does Weighting Actually Matter?

 ~12 min

Exercise 1 — Does Weighting Actually Matter?



The logic of survey weights

Surveys oversample some groups and undersample others. The weight variable corrects for this by telling R how many people each respondent represents in the real population.

A weight of 2.0 means 'count me twice'; 0.5 means 'count me half.' If we ignore weights, our estimates are biased toward whoever happened to respond — typically urban, educated, older respondents.



Your tasks

1. Compute unweighted and weighted mean democracy satisfaction side by side for each country
2. Find the country where weighting changes the estimate the most
3. Visualise the comparison as a scatter plot where the diagonal = no difference

Key function:

```
weighted.mean(dem_sat_r, w = weight)
```

Ex 2

Who Trusts Their Democracy?

 ~15 min

Exercise 2 — Who Trusts Their Democracy?



Democracy satisfaction across Europe

Satisfaction with how democracy works varies enormously across European countries. Sweden consistently leads; Greece and France are at the bottom. But these country averages mask individual variation.

Do men and women evaluate democracy differently? Political science research suggests gender gaps in political trust exist in some contexts but not others — and the direction is not always consistent.



Your tasks

1. Build a ranked bar chart of weighted % satisfied per country — include 95% confidence interval error bars
2. Break the chart down by gender: does the gender gap look the same everywhere?

Key function:

```
survey_mean(satisfied,  
            vartype = "se") * 100
```

Ex 3

The Economic Mood

 ~15 min

Exercise 3 — The Economic Mood

Current economy vs. intergenerational pessimism

Economic attitudes come in two flavours: how people feel about the economy right now, and how they feel about the future their children will inherit.

These two measures often diverge. A country may have relatively high economic satisfaction today (low unemployment, stable growth) while still believing the social contract is broken — that the next generation will not do as well as their parents.

Your tasks

1. Compute two indicators per country: % rating the economy as good + % saying children will be worse off
2. Visualise them side by side as a grouped bar chart
3. Identify the country with the largest gap between the two indicators

Key function:

```
pivot_longer(cols = c(pct_econ_good,  
  pct_pessimist))
```

Ex 4

Religion and Tolerance

 ~15 min

Exercise 4 — Religion and Tolerance



Religiosity and social acceptance

The relationship between religiosity and social conservatism is one of the most replicated findings in comparative political sociology. More religious populations tend to hold more traditional views on gender roles, sexuality, and minority rights.

But the relationship is not linear everywhere. Economic and historical context matters — and individual countries often defy the trend for reasons rooted in their specific national histories.



Your tasks

1. Compute % for whom religion is 'very important' and % comfortable with a gay child per country
2. Make a scatter plot with one dot per country and a linear trend line
3. Identify outliers — countries that don't fit the trend — and try to explain them

Key function:

```
geom_smooth(method = "lm",  
            se = FALSE)
```

Ex 5

Who Wants a Strong Leader?

 ~15 min

Exercise 5 — Who Wants a Strong Leader?



Support for autocracy in Europe

The Pew survey asked respondents to rate different ways of governing — including a system where 'a strong leader can make decisions without interference from parliament or the courts.'

This question taps into authoritarian predispositions. Support is not evenly distributed: it tends to be higher among older respondents, those on the ideological right, and those who are already dissatisfied with how democracy is working.

Note: this question was only asked in Hungary and Poland in our subset.



Your tasks

1. Check which countries have data for the political systems questions
2. Compare support for autocracy vs. representative democracy side by side
3. Split by left/centre/right ideology using `case_when()` — does ideology predict autocracy support?

Key function:

```
case_when(  
  ideology <= 2 ~ "Left",  
  ideology >= 4 ~ "Right",  
  TRUE ~ "Centre")
```

Ex 6

Your Group's Question

 ~18 min

Exercise 6 — Your Group's Question

Open exploration

This exercise has no pre-written code and no fixed answer. Your group formulates a research question, builds the analysis, and prepares to share one finding.

Starting points:

- Does misinformation concern vary with democracy dissatisfaction?
- Is support for gender equality related to religiosity?
- Does political ideology predict economic pessimism?
- Choose any two variables from the reference table in the document.

The goal is not a perfect result — it's a defensible finding you can explain in 2 minutes.

Your tasks

1. Agree on a research question — write it as a comment in your code chunk
2. Build the analysis using the scaffold provided in the .qmd file
3. Write 2–3 sentences of interpretation
4. Prepare to share one finding with the class

Key function:

```
# Your code here
```



Switch Roles Now

Navigator → Driver · Driver → Navigator

New Navigator: look back at Exercises 1–3 and explain to your Driver — what does `survey_mean()` do, and why do we use `pew_svy` instead of `pew_clean`?



Share Your Findings

2–3 minutes per group · every group shares one thing

1

What did you find?

Show the plot or table. State the result in one sentence.

2

Why is it surprising (or not)?

Compare it to what you expected before running the code.

3

What would you need to explain it better?

A different variable? A longer time series? An interview?

Good work today.

Three things your Driver should now be able to explain from memory:

- 1 Why we use `pew_svy` and not `pew_clean` for all weighted estimates
- 2 What `survey_mean(x, vartype = "se") * 100` returns — and why the SE is already in percentage points
- 3 Why `pivot_longer()` is needed before making grouped bar charts